RIZAL PEST CONTROL COMPANY

(RIZPECON)

A Project

Presented to the

Faculty of College of Engineering and Digital Technology

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V.V. Soliven Avenue II Cainta Rizal

Bachelor of Science Major in Computer Engineering

BY.

BRANDON ASHLEY P.

CADIMAS JULLIANA E.

DELOS SANTOS EMMANUEL B.

ESCOBAL RUBEN III M.

ESTARIS KRISTINE YVONNE F.

FRANCISCO JENNICA HAZEL V.

HERRADURA LAMECH UZ R.

MACABUDBOD CLYDE L.

VALDEZ KENNETH JAMES M.

INTRODUCTION

In today's modern world, various types of digital thermometers have been developed with different designs and added features to improve health monitoring. These devices play an essential role in detecting body temperature, particularly in identifying fever and other health conditions. However, most conventional digital thermometers are limited to manual and time-based measurements. This requires users—especially parents or guardians—to repeatedly check a child's temperature, which can be inconvenient and inefficient while performing daily tasks such as household chores. Continuous temperature monitoring is crucial for ensuring the safety and well-being of children. Sudden changes in body temperature, particularly high fever, may indicate serious health concerns that require immediate attention. Unfortunately, many existing thermometers do not provide real-time monitoring or automatic alerts, which may delay necessary action. To address these gaps, the researchers propose ThermoHead, a wearable headphone-style device designed to function as a continuous body temperature monitoring system. Unlike traditional thermometers, ThermoHead allows users to observe temperature changes in real time without interrupting their activities. The device features built-in sensors, a digital display showing LOW, MID, and HIGH temperature ranges using color indicators, and smart functions like notifications and suggested actions based on the user's condition.

The wearable design promotes comfort, ease of use, and accessibility, making it suitable for everyday use. As wearable health technology continues to expand, ThermoHead contributes to this growing field by combining convenience, functionality, and modern design. Overall, this study aims to develop an

innovative, practical, and user-friendly solution that enhances health awareness, improves response time to fever-related conditions, and supports caregivers in maintaining the well-being of children.

OBJECTIVES

Main Objective

To design and develop a wearable, child-safe, rechargeable device that continuously monitors body temperature and provides real-time alerts and guidance to caregivers.

Specific Objectives

1. To measure and record the child's temperature continuously using a non-contact IR sensor positioned at the temple area.
2. To classify temperature readings into Low, Mid, and High fever levels and visually display the status using an OLED screen and LED indicators.
3. To provide real-time alerts through LEDs, on-screen warnings, and Bluetooth notifications sent to a paired mobile device.
4. To offer suggested actions based on the detected temperature level to assist caregivers in responding appropriately.
5. To implement a rechargeable power system with built-in protection, auto-cutoff, and low-power operation to support several hours of continuous use.
6. To enhance comfort and usability through an adjustable, child-friendly headphone-style wearable design.
7. To incorporate safety features such as auto sleep mode and overheating protection to ensure safe and reliable device operation.

STATEMENT OF THE PROBLEM

Despite the availability of digital thermometers, most existing devices still rely on manual and occasional temperature checks, making it difficult for parents and caregivers to monitor a child continuously. Traditional thermometers do not offer real-time tracking, automated alerts, or guidance when a child's temperature rises suddenly. As a result, caregivers may fail to notice dangerous fever spikes, which can lead to delayed intervention and increased health risks. Therefore, there is a need for a child-friendly, wearable, and real-time temperature monitoring device that provides continuous measurements, quick alerts, and actionable recommendations to help ensure the safety and well-being of children.

COMPONENTS

HARDWARE	FUNCTION

COMPONENTS